



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Alternate Assessment Tool

Course Title: INTERNET OF THINGS AND 5G

Course Code: 22AI642

Semester: VI

Submitted by

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2025-26**

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Activity 1: TinkerCAD Aurdino

Problem Statement	Traditional street lighting systems operate continuously regardless of environmental conditions, leading to unnecessary energy consumption and increased costs. There is a need for a low-cost automated system that can intelligently control street lights based on ambient light conditions, ensuring energy efficiency and reducing power wastage.				
AAT MARKS	Design (5)	Execution (5)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Tinker Aurdino Uno				
MAP SDG	- SDG 7 – Affordable and Clean Energy - SDG 11 – Sustainable Cities and Communities - SDG 13 – Climate Action				
MAP COURSE OUTCOME of IOT/ BFE	CO	Mapping	Justification		
	CO1	Yes	Demonstrates perception layer of IoT using temperature/fire sensor, microcontroller processes input, and buzzer acts as output		
	CO2	Yes	Uses Arduino programming to design an automated fire detection and alert system		
	CO3	No	Does not involve advanced data analytics or storage; focuses only on real-time detection		
	CO4	Yes	Implements real-time embedded system for immediate alert generation using sensor-actuator integration		
MAP PROGRAM OUTCOME	PO	Mapping	Justification		
	PO1	Yes	Applied knowledge of electronics (temperature/fire sensors), basic science, and embedded programming for detecting fire conditions		
	PO2	Yes	Identified and analysed the real-world problem of delayed fire detection and the need for faster response systems		
	PO3	Yes	Designed an automated IoT-based fire alarm system to		

			provide early detection and instant alerts
	PO5	Yes	Used modern tools such as Arduino IDE and simulation platforms (e.g., Tinkercad) for system design and implementation
	PO6	Yes	Addresses societal safety by preventing fire hazards, reducing damage, and protecting lives
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	PSO	Mapping	Justification
	PSO1	Yes	Demonstrates programming and problem-solving skills by implementing sensor-based fire detection logic with threshold conditions and alert mechanisms
	PSO2	Yes	Identifies and solves the real-world problem of fire hazard detection by developing an automated system for early warning and safety
	PSO3	Yes	Utilizes simulation tools like Tinkercad or embedded platforms to design, test, and deploy the smart fire alarm system
URL Link	https://www.tinkercad.com/things/kk75jx30q4E-remote-patient-monitoring?sharecode=Q4jgyYgvNHrJhOHfSG0MwzJSLIfgrCOR06lQy8_gOkU		
SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model			

	<pre> graph TD Start([Start]) --> Setup[void setup() pinMode(led, OUTPUT); Serial.begin(9600);] Setup --> Loop[void loop() int value = analogRead(ldr); Serial.println(value);] Loop --> Decision{Is Value < 500?} Decision -- Yes --> OnLED[digitalWrite(I, HIGH); digitalWrite(led, HIGH);] Decision -- No --> OffLED[Turn OFF LED digitalWrite(led, LOW);] OnLED --> Delay[delay(200);] OffLED --> Delay Delay --> End([End]) </pre>
COURSE COORDINATOR	Dr ARUNA M G

Details

Abstract:

The Smart Street Lighting System is an IoT-based solution designed to automatically control street lights based on ambient light conditions. The system utilizes a Light Dependent Resistor (LDR) sensor connected to an Arduino UNO microcontroller to detect the intensity of surrounding light. Based on the sensor readings, the system intelligently turns the street light ON during low light conditions (night) and OFF during high light conditions (day), ensuring efficient energy utilization.

The primary objective of this system is to reduce unnecessary power consumption and enhance energy efficiency in public lighting infrastructure. By automating the operation of street lights, the system minimizes human intervention and operational costs while contributing to sustainable energy usage. This is particularly beneficial for urban and rural areas where continuous manual monitoring of street lights is impractical.

The proposed system demonstrates how embedded systems and sensor-based automation can be integrated to develop smart city solutions. It also has potential for future enhancements such as motion-based lighting, remote monitoring, and integration with IoT platforms, making it a scalable and environmentally friendly solution for modern infrastructure.

Use Case:

- Automatic switching of street lights based on day and night conditions
- Energy-efficient lighting in urban and rural areas
- Reduction of manual intervention in street light operation
- Smart city infrastructure development with sustainable lighting systems

Introduction:

With the rapid advancement of technology, urban infrastructure is increasingly adopting smart solutions to improve efficiency and sustainability. Traditional street lighting systems operate continuously without considering environmental conditions, leading to unnecessary energy consumption and increased operational costs. This inefficiency highlights the need for an automated and intelligent lighting system that can adapt to changing surroundings.

The Smart Street Lighting System is an Internet of Things (IoT)-based solution designed to address this issue by automatically controlling street lights based on ambient light conditions. The system uses a Light Dependent Resistor (LDR) sensor connected to an Arduino UNO microcontroller to detect light intensity. Based on the sensor data, the system processes the input and turns the street light ON during low light conditions (night) and OFF during high light conditions (day), ensuring efficient energy usage.

The primary goal of this project is to develop a low-cost, reliable, and energy-efficient solution for automated street lighting. This system reduces manual intervention, minimizes power wastage, and supports the development of smart city infrastructure. By integrating embedded systems with sensor-based automation, the project demonstrates how modern technology can be utilized to create sustainable and scalable solutions. Future enhancements may include motion-based lighting, remote monitoring, and integration with IoT platforms for advanced smart city applications.

Objective:

- To develop a Smart Street Lighting System using Arduino UNO
- To detect ambient light intensity using an LDR sensor
- To automatically control street lights based on day and night conditions
- To reduce energy consumption and improve efficiency through automation

General procedure:

- Design and connect the circuit using Arduino UNO, LDR sensor, LED, and resistor in Tinkercad
- Develop and upload code to read ambient light intensity from the LDR sensor
- Continuously monitor light levels in real time
- Compare sensor values with predefined threshold to determine day or night condition
- Automatically switch the street light ON during low light and OFF during high light conditions

Step to Execute:

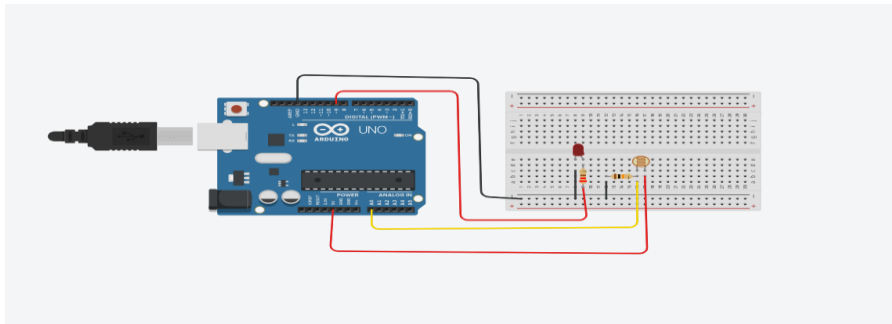
- Set up the circuit in Tinkercad and upload the Arduino code
- Start the simulation/system to begin reading light intensity from the LDR sensor
- Observe the sensor values in the Serial Monitor
- Verify automatic switching of the LED based on light conditions (day/night)

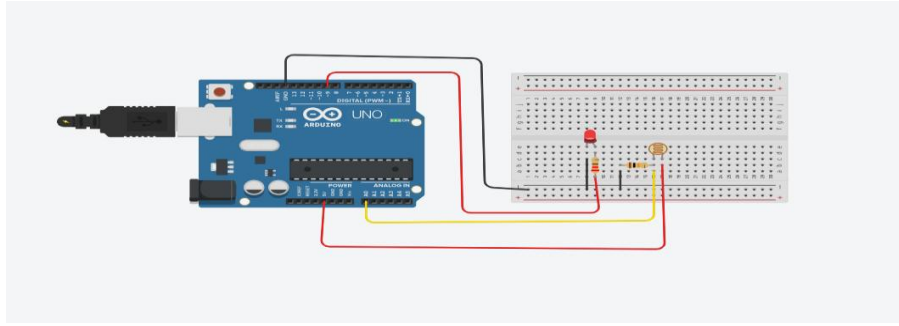
Source Code:

```
int ldr = A0;
int led = 9;
void setup() {
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  int value = analogRead(ldr);
  Serial.println(value);
  if (value < 500) { // Dark condition
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
  delay(200);
}
```

Output:

- Real-time monitoring of ambient light intensity using LDR sensor
- Automatic switching of street light (LED) based on day and night conditions
- Display of sensor values in the Serial Monitor
- Continuous operation ensuring energy-efficient lighting system

Photos/ Screenshot/Snapshot :

**Outcome:**

- Successfully developed a Smart Street Lighting system prototype using Arduino UNO
- Achieved automatic control of street lights based on ambient light conditions
- Reduced energy consumption through efficient and intelligent lighting control
- Eliminated the need for manual operation of street lights
- Demonstrated a low-cost and scalable solution for smart city infrastructure

Activity 2: Wokwi ESP32 and THINGSPEAK

Problem Statement	Traditional security systems often lack real-time motion detection and require continuous human monitoring, which can lead to delayed response to unauthorized activities. There is a need for a low-cost, automated system that can detect human movement instantly, provide immediate alerts, and enhance safety in homes, offices, and restricted areas.				
AAT MARKS	Design (5)	Execution (5)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Wokwi ESP and THINGSPEAK				
MAP SDG	- SDG 11: Sustainable Cities and Communities - SDG 16: Peace, Justice and Strong Institutions				
MAP COURSE OUTCOME of IOT/BFE	CO	Mapping	Justification		
	CO1	Yes	Demonstrates basic IoT architecture including sensing (PIR), processing (ESP32), and actuation (LED/alert system)		
	CO2	No	Does not involve complex automation or multi-device IoT system design; limited to single-sensor motion detection		
	CO3	Yes	Applies data acquisition and processing by detecting motion using PIR sensor and generating real-time output		
	CO4	Yes	Demonstrates working of sensors and actuators through integration of PIR sensor and LED in embedded system		
MAP PROGRAM OUTCOME	PO	Mapping	Justification		
	PO1	Yes	Applies knowledge of embedded systems, PIR sensors, and programming for motion detection		
	PO2	Yes	Identifies and analyzes the real-world issue of delayed response in traditional security systems		
	PO3	Yes	Designs an automated motion detection system for instant alert generation		
	PO5	Yes	Uses modern tools like Wokwi ESP simulator and		

			ThingSpeak cloud platform for implementation and monitoring
	PO6	Yes	Enhances safety and security in homes and restricted areas by enabling real-time detection
	PO4	No	Does not involve detailed investigation, experimentation, or data-driven analysis
	PO7	No	Does not address environmental sustainability or energy optimization aspects
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	PSO	Mapping	Justification
	PSO1	Yes	Demonstrates programming and problem-solving by implementing PIR-based motion detection logic
	PSO2	No	Does not apply advanced AI/ML techniques; uses simple threshold-based detection logic
	PSO3	Yes	Utilizes cloud platform (ThingSpeak) to monitor system activity, demonstrating basic cloud resource usage
URL Link	https://wokwi.com/projects/460820904371564545		
SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model			

	<pre>graph TD; PIR[PIR Sensor] -- Digital Signal --> ESP32[ESP32 Processor]; ESP32 -- Control Signal --> LED1[LED]; ESP32 -- LED --> LED2[LED]; ESP32 -- Serial Output --> SM[Serial Monitor];</pre>
COURSE COORDINATOR	Dr ARUNA M G

Details

Abstract:

Security and surveillance are essential for ensuring safety in homes, offices, and public spaces. This project presents a smart motion detection system using ESP32 and a PIR (Passive Infrared) sensor to detect human movement in real time. The system processes sensor input and identifies the presence or absence of motion. When motion is detected, the system activates an LED as a visual alert and displays status messages through the Serial Monitor.

The primary objective of this system is to provide a simple, low-cost, and efficient solution for motion-based monitoring. It can be used for intrusion detection, automatic lighting, and basic security applications. The system demonstrates how embedded systems and sensors can be integrated to develop intelligent monitoring solutions. Future enhancements may include buzzer alerts, camera integration, and IoT-based remote notifications, making it a scalable and effective solution for modern security systems.

Use Case:

- **Home Security:** Detect unauthorized movement in homes and trigger alerts
- **Office Monitoring:** Monitor restricted areas and ensure workplace safety
- **Automatic Lighting Systems:** Turn lights ON/OFF based on human presence
- **Surveillance Systems:** Support smart security and monitoring applications in buildings and public spaces

Introduction:

Security and safety have become increasingly important in modern living environments, especially with the rise in unauthorized access and theft. Continuous monitoring of spaces is essential to detect movement and respond quickly to potential threats. Traditional security systems often require constant human supervision and can be costly, making them less accessible for everyday use.

This project presents a low-cost, IoT-based smart motion detection system using an ESP32 microcontroller and a PIR (Passive Infrared) sensor. The system detects human movement in real time by sensing changes in infrared radiation. Based on the sensor input, the ESP32 processes the data and

activates an LED to indicate motion detection, while also displaying status messages through the Serial Monitor.

The primary objective of this system is to provide a simple, efficient, and cost-effective solution for basic security and automation applications. It demonstrates how embedded systems and sensors can be integrated to build smart monitoring solutions. Future enhancements may include buzzer alerts, camera integration, and cloud-based remote monitoring, making the system more advanced and scalable for real-world applications.

Objective:

- To design and develop a low-cost motion detection system using ESP32
- To detect human movement in real time using a PIR sensor
- To provide visual indication through an LED when motion is detected
- To implement a simple and efficient system for security and automation applications

General procedure:

- Connect the PIR sensor, LED, and resistor to the ESP32 in Wokwi
- Initialize the PIR sensor and LED using MicroPython code
- Continuously read motion data from the PIR sensor
- Process the sensor input to detect presence or absence of motion
- Turn ON the LED when motion is detected and OFF otherwise
- Display motion status in the Serial Monitor for real-time monitoring

Step to Execute:

- **Setup Circuit:** Connect ESP32 with PIR sensor, LED, and resistor in Wokwi
- **Upload Code:** Write and upload MicroPython code to ESP32
- **Initialize System:** Configure PIR sensor and LED pins
- **Read Sensor Data:** Continuously read motion signals from the PIR sensor
- **Process Data:** Detect presence or absence of motion based on sensor input
- **Display Output:** Show motion status in the Serial Monitor
- **Trigger Output:** Turn ON LED when motion is detected and OFF otherwise.

Source code:

```
from machine import Pin
import time

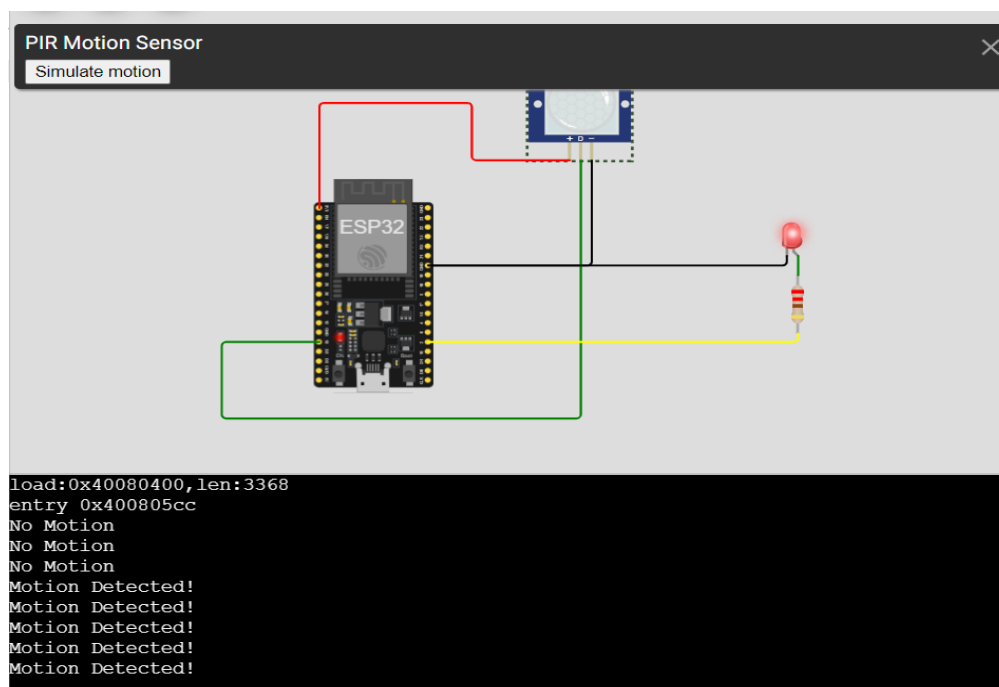
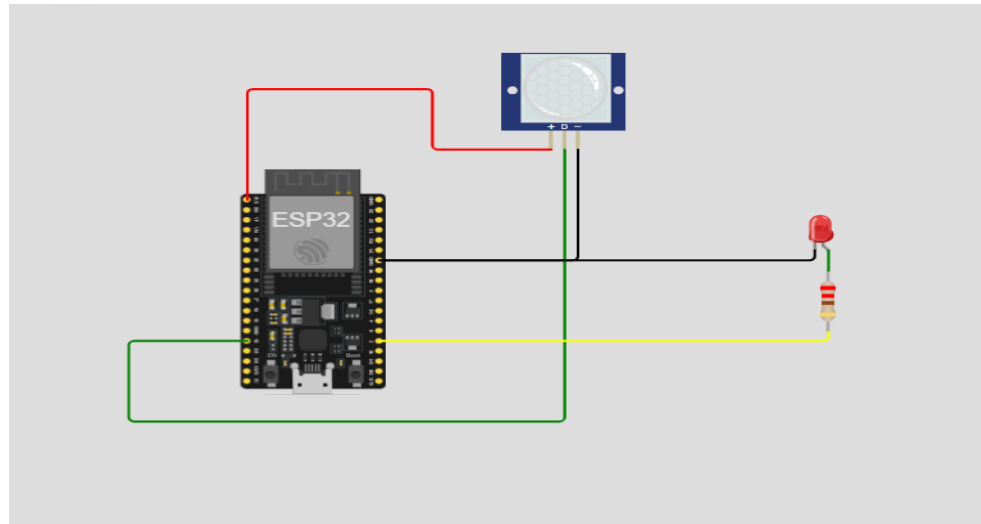
pir = Pin(13, Pin.IN)
led = Pin(2, Pin.OUT)

while True:
    if pir.value() == 1:
        led.value(1)
        print("Motion Detected!")
    else:
        led.value(0)
        print("No Motion")
    time.sleep(0.5)
```

Output:

- The system successfully detects human motion using the PIR sensor
- Real-time motion status is displayed in the Serial Monitor
- LED turns ON when motion is detected and OFF when no motion is present
- Continuous monitoring of movement ensures effective basic security system

Photos/ Screenshot/ Snapshot:



Outcome:

The project successfully demonstrates a low-cost IoT-based motion detection system that can detect human movement in real time and provide immediate visual indication through an LED. It ensures continuous monitoring of surroundings and enhances basic security by identifying motion instantly, thereby contributing to safer and smarter environments.

Activity 3: Wokwi STM

Problem Statement	Traditional lighting systems lack efficient and precise control over brightness levels, leading to energy wastage and limited flexibility in lighting applications. There is a need for a simple and cost-effective system that can adjust light intensity in real time based on user input, ensuring better energy efficiency and improved user control.				
AAT MARKS	Design (5)	Execution (5)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Wokwi STM				
MAP SDG	- SDG 7: Affordable and Clean Energy - SDG 9: Industry, Innovation and Infrastructure				
MAP COURSE OUTCOME of IOT/BFE	CO	Mapping	Justification		
	CO1	Yes	Demonstrates basic IoT/embedded system architecture involving input (potentiometer), processing (microcontroller), and output (LED brightness control)		
	CO2	Yes	Applies programming and control techniques (PWM) to regulate LED brightness based on user input		
	CO3	No	Does not involve complex data acquisition, storage, or processing; only real-time input-output control is performed		
	CO4	Yes	Demonstrates working of sensors (potentiometer) and actuators (LED) for real-time brightness adjustment		
MAP PROGRAM OUTCOME	PO	Mapping	Justification		
	PO1	Yes	Applies knowledge of embedded systems, analog input, and PWM techniques for brightness control		
	PO2	Yes	Identifies the issue of inefficient lighting and lack of control in traditional systems		
	PO3	Yes	Designs a PWM-based LED control system for efficient and flexible lighting		
	PO5	Yes	Uses Wokwi STM simulator and embedded programming		

			tools for system development and testing
	PO6	Yes	Contributes to energy efficiency by reducing power wastage through controlled lighting
	PO4	No	Does not involve experimental investigation or performance analysis of multiple design approaches
	PO7	Yes	Supports energy conservation aligning with sustainable energy practices
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	PSO	Mapping	Justification
	PSO1	Yes	Demonstrates programming and problem-solving through PWM-based LED control using real-time input
	PSO2	No	Does not involve AI/ML-based intelligent decision-making; relies on direct user input
	PSO3	Yes	Utilizes simulation platform (Wokwi STM) to design, test, and validate the system
URL Link	https://wokwi.com/projects/460830329368907777		
SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model	SYSTEM LED Brightness Control System <pre> graph TD Start([Start]) --> Init[Initialize STM32, Potentiometer & LED] Init --> Read{{Read Potentiometer Value (0-1023)}} Read --> Map{{Map Value to PWM Range (0-255)}} Map --> Adjust[Adjust LED Brightness] Adjust --> End([End]) </pre>		
COURSE COORDINATOR	Dr ARUNA M G		

Details
Abstract: Efficient water management is essential in agriculture to improve crop yield and reduce resource wastage. This project presents a Smart Irrigation System using an STM32 microcontroller to automate the irrigation process based on soil moisture levels. The system uses a soil moisture sensor to continuously monitor the water content in the soil and processes the data in real time. Based on predefined threshold values, the system automatically controls a water pump to supply water only when required. Real-time information is displayed on an LCD screen, and a buzzer provides alerts when the soil becomes excessively dry. This automation reduces manual intervention, prevents over-irrigation, and ensures optimal water usage. The proposed system demonstrates a low-cost, efficient, and scalable solution for modern agriculture. Future enhancements may include IoT-based remote monitoring, weather integration, and intelligent irrigation scheduling for improved productivity and sustainability.
Use Case: • Automatic Irrigation: The system automatically turns the water pump ON/OFF based on soil moisture levels, reducing manual intervention. • Water Conservation: Prevents over-irrigation by supplying water only when required, ensuring efficient use of water resources. • Smart Farming: Enables farmers to monitor and manage irrigation efficiently for better crop growth. • Agricultural Automation: Supports modern farming practices by integrating sensors and automated control systems.
Introduction: Agriculture plays a crucial role in the economy, and efficient water management is essential for improving crop productivity and sustainability. Traditional irrigation methods often result in excessive water usage and require constant manual supervision, which can lead to inefficiencies and resource wastage. This highlights the need for an automated system that can monitor soil conditions and manage irrigation effectively.

This project presents a Smart Irrigation System using an STM32 microcontroller to address these challenges. The system uses a soil moisture sensor to continuously monitor the moisture content in the soil and processes the data in real time. Based on the sensor readings, the system automatically controls a water pump to supply water only when necessary. It also provides real-time feedback through an LCD display and generates alerts using a buzzer when the soil becomes too dry.

The primary goal of this project is to develop a low-cost, efficient, and automated irrigation solution that reduces water wastage and minimizes human effort. By integrating sensors and embedded systems, the project demonstrates how modern technology can support sustainable and smart farming practices. Future enhancements may include IoT-based remote monitoring and intelligent irrigation scheduling for improved agricultural efficiency.

Objective:

- To design and develop an automated irrigation system using an STM32 microcontroller
- To monitor soil moisture levels in real time using a sensor
- To automatically control the water pump based on soil moisture conditions
- To reduce water wastage and improve irrigation efficiency through smart automation

General procedure:

- Initialize the STM32 microcontroller, LCD display, and input/output pins
- Read soil moisture data from the sensor (or potentiometer in simulation) using analog input
- Process the sensor data and compare it with predefined threshold values
- Display the moisture level and system status on the LCD
- Automatically turn the water pump ON or OFF based on soil moisture condition
- Activate the buzzer when the soil becomes too dry

Step to Execute:

- Set up the circuit in Wokwi by connecting the STM32 board, potentiometer (soil moisture sensor), LCD, LED (pump), and buzzer
- Upload the program code to the STM32 microcontroller
- Start the simulation in Wokwi
- Adjust the potentiometer to simulate different soil moisture levels

-
- Observe the LCD display for moisture values and system status
 - Verify that the pump (LED) and buzzer respond correctly based on the soil moisture condition

Source Code:

```
int pot = A0;
int led = A5;

void setup() {
  pinMode(led, OUTPUT);
}

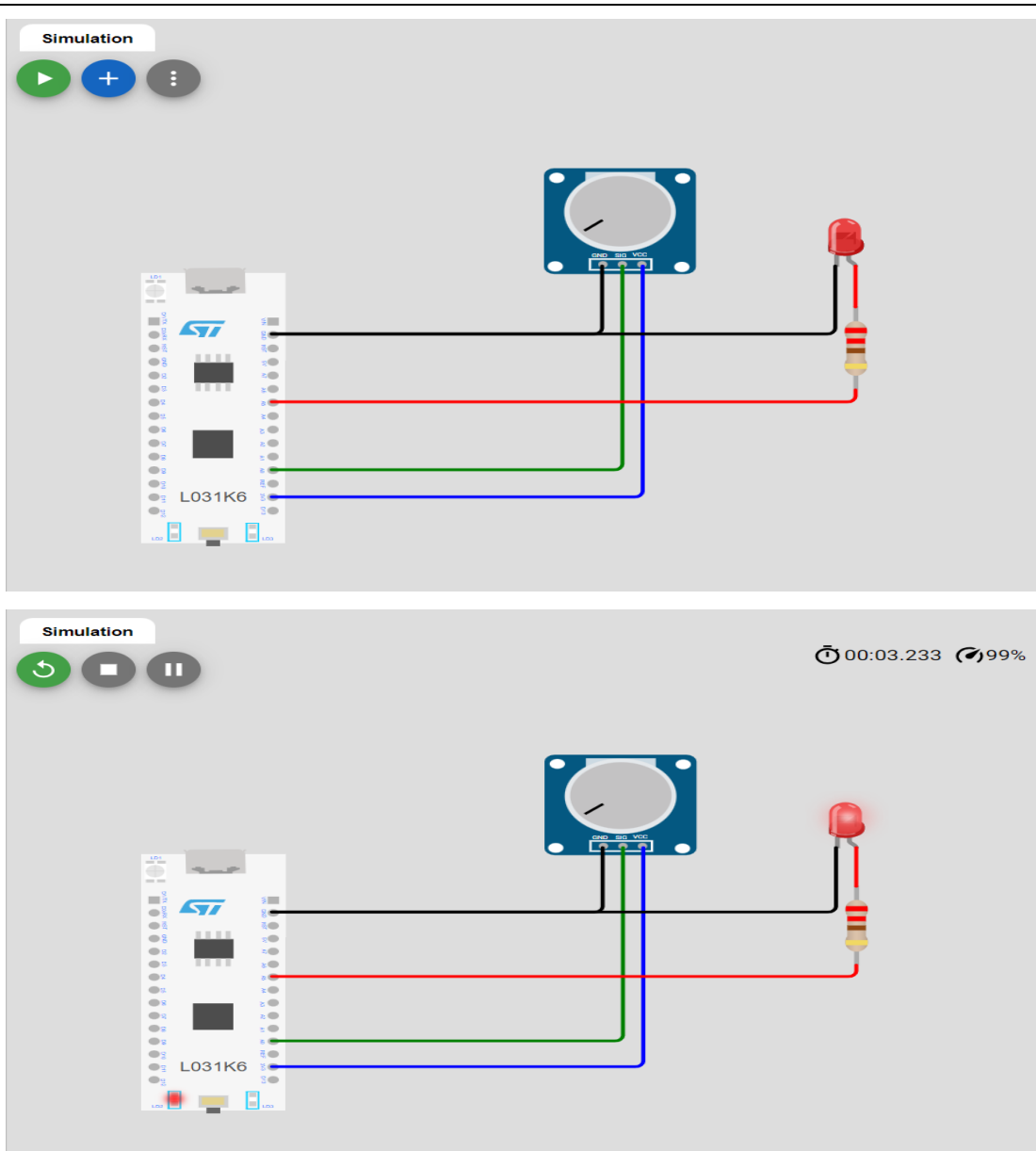
void loop() {
  int value = analogRead(pot); // 0–1023
  int brightness = map(value, 0, 1023, 0, 255);

  analogWrite(led, brightness);

  delay(10);
}
```

Output: The system successfully displays real-time soil moisture levels on the LCD and automatically controls the water pump based on predefined threshold values. When the soil is dry, the pump (LED) turns ON and the buzzer is activated, while under normal and wet conditions, the pump remains OFF. The system responds accurately to changes in moisture levels, demonstrating efficient and automated irrigation control.

Photos/ Screenshot/Snapshot :



Outcome: The project successfully demonstrates an efficient and automated irrigation system that optimizes water usage based on real-time soil moisture conditions. It reduces manual effort, prevents water wastage, and improves crop management. The system also provides a strong foundation for future enhancements such as IoT integration and AI-based smart farming solutions, making it scalable and suitable for modern agricultural practices.

Activity 4: Wokwi Raspberry Pi Pico Simulator

Problem Statement	Ensuring security in homes and restricted areas is challenging due to the need for constant monitoring and high-cost systems. A low-cost solution is required to automatically detect motion and provide instant alerts for improved safety.				
AAT MARKS	Design (5)	Execution (5)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Wokwi Raspberry Pi Pico Simulator				
MAP SDG	SDG 11: Sustainable Cities and Communities SDG 16: Peace, Justice and Strong Institutions				
MAP COURSE OUTCOME of IOT/BFE	CO	Mapping	Justification		
	CO1	Yes	Demonstrates IoT architecture by integrating motion sensor (input), Raspberry Pi Pico (processing), and buzzer/LED (output)		
	CO2	Yes	Applies programming and IoT concepts to implement motion-based theft detection using embedded system logic		
	CO3	No	Does not involve data storage, analytics, or advanced processing; system performs only real-time detection		
	CO4	Yes	Demonstrates working of sensors and actuators through motion detection and alert generation in real time		
MAP PROGRAM OUTCOME	PO	Mapping	Justification		
	PO1	Yes	Applies knowledge of embedded systems, motion sensors, and programming for theft detection		
	PO2	Yes	Identifies and analyzes the problem of lack of affordable and automated security systems		
	PO3	Yes	Designs an automated theft detection system for real-time monitoring and alert generation		
	PO5	Yes	Uses Wokwi Raspberry Pi Pico simulator and embedded programming for system implementation		

	PO6	Yes	Enhances safety and security in homes and restricted areas by providing instant alerts
	PO4	No	Does not involve experimental research, testing multiple models, or detailed performance analysis
	PO7	No	Does not directly address environmental or sustainability concerns
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	PSO	Mapping	Justification
	PSO1	Yes	Demonstrates programming and problem-solving through motion-based detection and alert system implementation
	PSO2	Yes	Solves real-world security problems using simple decision logic for motion detection
	PSO3	Yes	Utilizes simulation tools (Wokwi) to design, test, and validate the embedded system
	PSO4	No	Does not involve advanced AI/ML techniques or intelligent predictive models
URL Link	https://wokwi.com/projects/460796653093014529		

SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model	Smart Theft Detection System Using Raspberry Pi Pico <pre> graph TD Start([Start]) --> DetectMotion[Detect Motion] DetectMotion --> CheckIntrusion{Check for Intrusion} CheckIntrusion -- Yes --> ActivateAlarm[Activate Alarm] ActivateAlarm --> AlertUser[Alert User] AlertUser --> End([End]) AlertUser --> CheckIntrusion </pre>
COURSE COORDINATOR	Dr ARUNA M G

Details
Abstract: Security is a critical concern in homes and restricted areas, requiring reliable and real-time monitoring systems. This project presents a Smart Theft Detection System using a Raspberry Pi Pico that detects unauthorized movement through a PIR motion sensor. The system processes the sensor input and triggers alerts using a buzzer, LED, and LCD display to indicate intrusion instantly. The primary objective of this system is to provide a simple, low-cost, and efficient security solution. It enables real-time monitoring and immediate alert generation, reducing the need for constant human supervision. The project demonstrates how embedded systems and basic IoT concepts can be integrated to develop effective safety applications. Future enhancements may include wireless notifications and cloud-based monitoring for improved security and scalability.
Use Case: • Home security system for detecting intruders • Monitoring of offices and restricted areas • ATM and bank security alert system • Protection of warehouses and storage units • Night-time surveillance in buildings and campuses
Introduction: Security has become a major concern in homes, offices, and public spaces due to the increasing risk of unauthorized access and theft. Traditional security systems are often costly and require continuous human monitoring, making them less practical for everyday use. This creates a need for an automated and efficient system that can detect intrusions in real time and provide immediate alerts. This project presents a Smart Theft Detection System using a Raspberry Pi Pico that automatically detects motion through a PIR sensor. The system processes the sensor input and triggers alerts using an LED, buzzer, and LCD display to indicate intrusion. It ensures real-time monitoring and instant response without human intervention.

The main objective of this system is to provide a simple, low-cost, and reliable security solution. By integrating sensors and embedded systems, the project demonstrates how modern technology can be used for effective safety applications. Future enhancements may include wireless notifications and IoT-based monitoring for improved security and scalability.

Objective:

- To design and develop a smart and cost-effective theft detection system
- To detect unauthorized motion using a PIR sensor in real time
- To trigger alerts using LED, buzzer, and LCD display upon intrusion
- To implement continuous real-time monitoring using Raspberry Pi Pico

General procedure:

- Connect the PIR sensor, LED, buzzer, and LCD to the Raspberry Pi Pico
- Initialize the system and configure all components using MicroPython
- Continuously monitor motion using the PIR sensor
- When motion is detected, activate the LED and buzzer, and display an alert message on the LCD
- When no motion is detected, keep the system in monitoring (safe) state

Step to Execute:

- Create a new Raspberry Pi Pico project in Wokwi
- Add and connect PIR sensor, LED, buzzer, and LCD components
- Write and upload the MicroPython code in main.py
- Run the simulation and verify proper component connections
- Trigger the PIR sensor to test motion detection and system response

Source Code:

```
from machine import Pin, ADC
import time

sensor = ADC(26)
led = Pin(15, Pin.OUT)
```

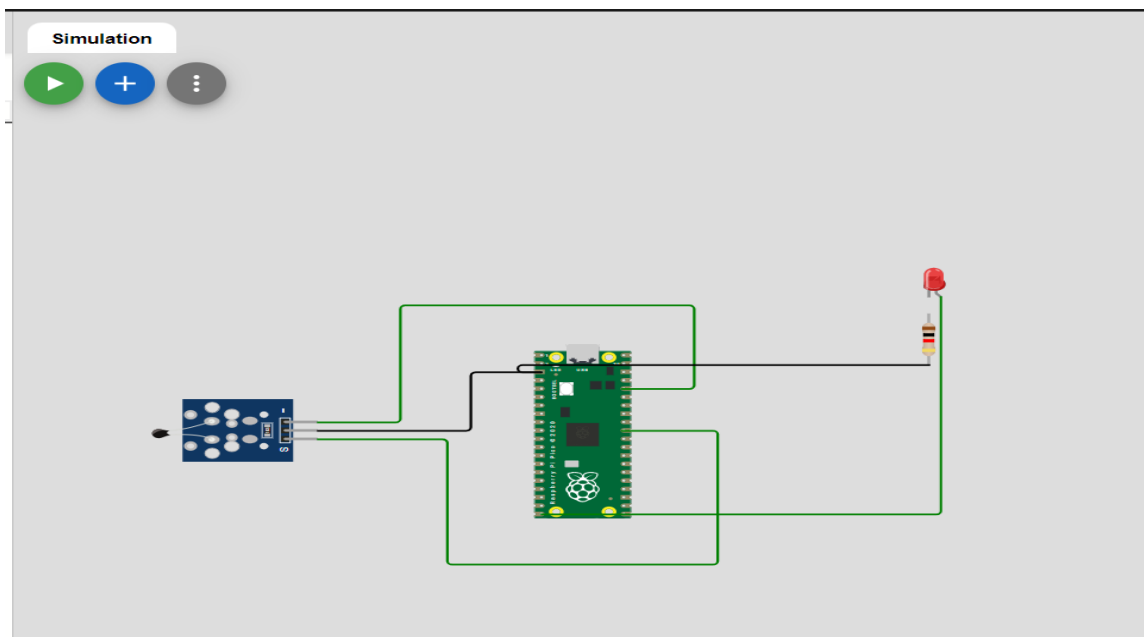
```
while True:
    value = sensor.read_u16()
    print("Value:", value)

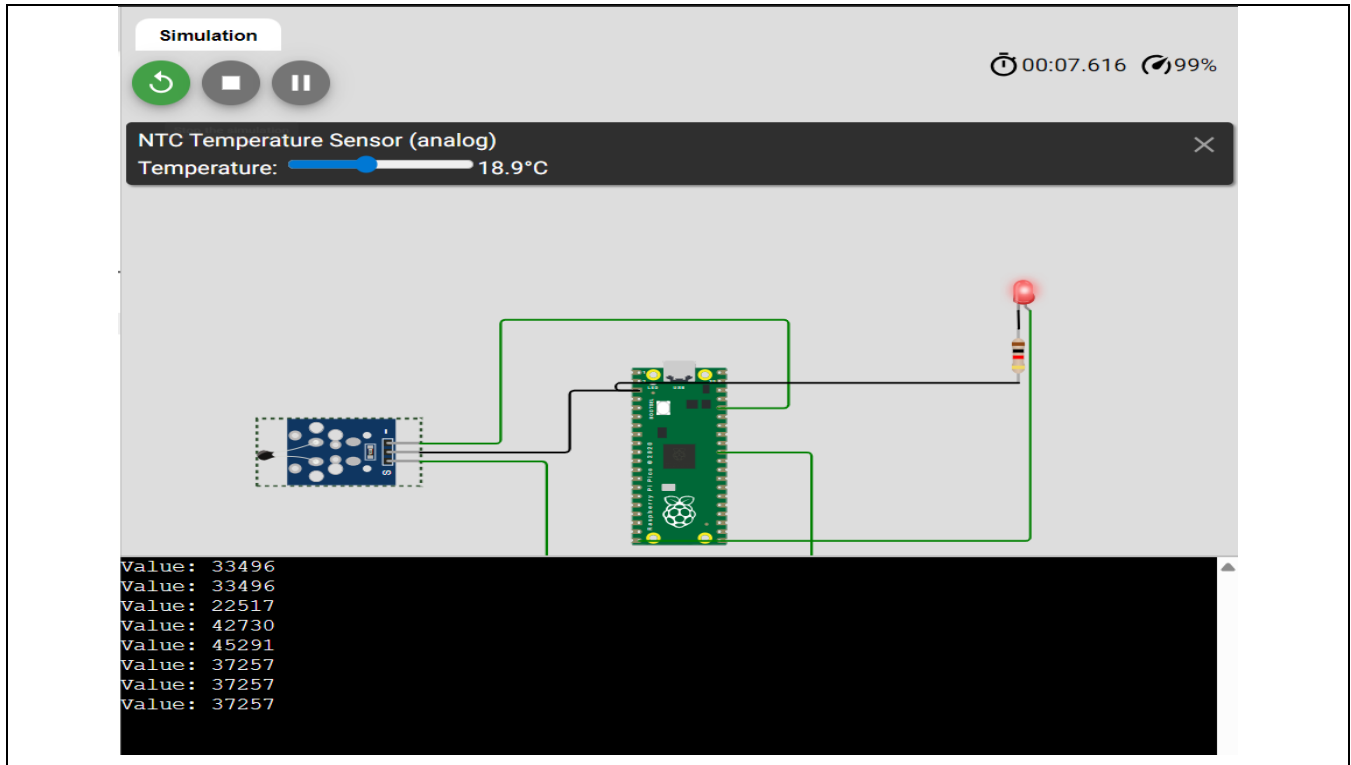
    if value > 30000:
        led.value(1)
    else:
        led.value(0)

    time.sleep(1)
```

Output:

- The system displays “Safe Area” when no motion is detected
- Upon motion detection, the LED and buzzer are activated
- The LCD displays an alert message “Motion Detected”
- The Serial Monitor shows real-time status of the system

Photos/ Screenshot/Snapshot :

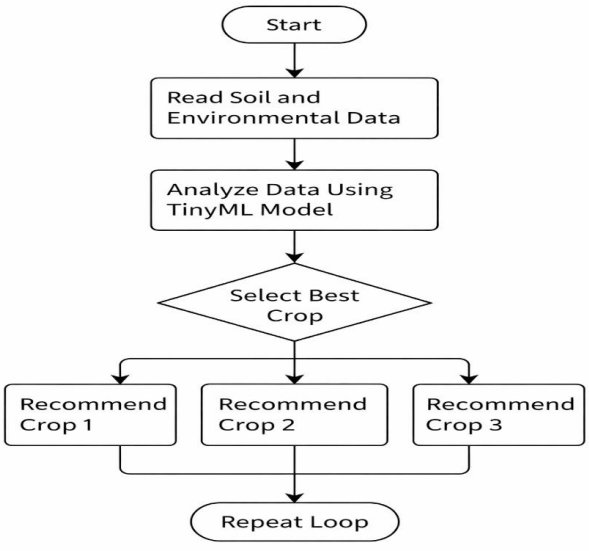


Outcome:

- Successfully developed a functional smart theft detection system
- Achieved accurate motion detection and reliable alert mechanisms
- Demonstrated effective integration of sensors and actuators using Raspberry Pi Pico
- Provided a simple, low-cost, and efficient solution for real-time security monitoring

Activity 5: TinyML simulation in Wokwi

Problem Statement	Farmers face difficulty in selecting suitable crops due to varying soil and environmental conditions. A smart, low-cost system is needed to analyze these factors and recommend the best crop in real time using TinyML.				
AAT MARKS	Design (5)	Execution (5)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Wokwi and Google colab				
MAP SDG	• SDG 2: Zero Hunger • SDG 12: Responsible Consumption and Production				
MAP COURSE OUTCOME of IOT/BFE	CO	Mapping	Justification		
	CO1	Yes	Demonstrates IoT system architecture with motion sensor (input), Raspberry Pi Pico (processing), and buzzer/LED (output)		
	CO2	Yes	Applies programming and IoT concepts to develop a motion-based theft detection system		
	CO3	No	Does not involve data analytics, storage, or advanced data processing; focuses only on real-time detection		
	CO4	Yes	Demonstrates working of sensors and actuators for real-time monitoring and alert generation		
MAP PROGRAM OUTCOME	PO	Mapping	Justification		
	PO1	Yes	Applies knowledge of embedded systems, motion sensors, and programming for theft detection		
	PO2	Yes	Identifies and analyzes the problem of lack of affordable and automated security systems		
	PO3	Yes	Designs an automated motion-based theft detection system for real-time alerts		
	PO5	Yes	Uses Wokwi Raspberry Pi Pico simulator and embedded programming tools for implementation		

	PO6	Yes	Enhances safety and security in homes and restricted areas through automated monitoring
	PO4	No	Does not involve detailed investigation, experimentation, or comparative analysis
	PO7	No	Does not address environmental or sustainability aspects
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	PSO	Mapping	Justification
	PSO1	Yes	Demonstrates programming and problem-solving through motion-based detection and alert system implementation
	PSO2	Yes	Solves real-world security problems using simple decision logic for motion detection
	PSO3	Yes	Utilizes simulation tools (Wokwi) to design, test, and validate the embedded system
	PSO4	No	Does not involve advanced AI/ML techniques or intelligent predictive models
URL Link	https://wokwi.com/projects/461081958618849281		
SYSTEM DIAGRAM / UML / SEQUENCE DIAGRAM/ 3D Model	 <pre> graph TD Start([Start]) --> Read[Read Soil and Environmental Data] Read --> Analyze[Analyze Data Using TinyML Model] Analyze --> Select{Select Best Crop} Select --> Crop1[Recommend Crop 1] Select --> Crop2[Recommend Crop 2] Select --> Crop3[Recommend Crop 3] Crop1 --> Repeat([Repeat Loop]) Crop2 --> Repeat Crop3 --> Repeat </pre>		

COURSE COORDINATOR	Dr ARUNA M G
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Details
Abstract: Efficient crop selection is essential for improving agricultural productivity and sustainability. This project presents a TinyML-based crop recommendation system that assists farmers in selecting the most suitable crops based on environmental and soil parameters. Key inputs such as nutrient levels and humidity are collected and processed using a trained machine learning model deployed on an ESP32 microcontroller. The system performs real-time predictions at the edge and displays the recommended crop on an LCD interface. By enabling data-driven decision-making without relying on cloud connectivity, the solution reduces latency and improves efficiency. This low-cost and scalable system demonstrates the potential of TinyML in smart agriculture, promoting sustainable farming practices and reducing dependency on manual analysis. Future enhancements may include integration with IoT platforms and additional environmental parameters for improved accuracy.
Use Case: • Crop selection assistance for farmers based on soil and environmental conditions • Smart farming support through real-time, data-driven decision making • Educational tool to demonstrate TinyML and IoT concepts for students and researchers
Introduction: Agriculture plays a vital role in the economy, yet selecting the right crop based on soil and environmental conditions remains a significant challenge for farmers. Traditional methods often rely on experience and intuition, which can lead to inefficient resource utilization and reduced crop yield. With the advancement of IoT and machine learning technologies, it is now possible to make intelligent and data-driven decisions in agriculture. This project presents a TinyML-based crop recommendation system that analyzes key parameters such as soil nutrients and humidity using an embedded device. The collected data is processed in real time using a

trained machine learning model deployed on the device. Based on the analysis, the system recommends the most suitable crop and displays the result on an LCD interface.

The primary objective of this system is to provide a low-cost, efficient, and real-time solution for crop selection. By enabling on-device intelligence, the system reduces dependency on cloud infrastructure and supports faster decision-making. This approach promotes sustainable farming practices and demonstrates the potential of TinyML in modern agriculture. Future enhancements may include additional parameters and IoT-based monitoring for improved accuracy and scalability.

Objective:

- To develop a smart crop recommendation system using TinyML
- To analyze soil and environmental parameters for accurate crop selection
- To implement real-time prediction using an ESP32 microcontroller

General procedure:

- Collect and preprocess crop dataset with parameters such as N, P, K, humidity, and temperature
- Train a machine learning model and define threshold-based classification (TinyML logic)
- Convert the model/logic into embedded-compatible code
- Interface ESP32 with potentiometer (input) and LEDs (output indicators)
- Read input values and process them using classification logic
- Indicate the predicted output (mode/crop) using LEDs in real time

Step to Execute:

- Prepare the dataset and design classification logic (TinyML/threshold-based model) for crop prediction
- Convert the trained model or logic into MicroPython/embedded code
- Set up the ESP32 with potentiometer (input) and LEDs (output indicators) in Wokwi
- Upload the code to the ESP32 simulation
- Run the simulation and vary input using the potentiometer
- Observe the predicted output (mode/crop) through LED indications

Source Code:

```
from machine import Pin, ADC
```

```
import time

led1 = Pin(2, Pin.OUT)
led2 = Pin(4, Pin.OUT)
led3 = Pin(5, Pin.OUT)

pot = ADC(Pin(34))
pot.atten(ADC.ATTN_11DB)

last_mode = None # to avoid repeated printing

while True:
    value = pot.read()

    # Turn OFF all LEDs
    led1.value(0)
    led2.value(0)
    led3.value(0)

    if value < 1200:
        mode = "MODE 1"
        led1.value(1)

    elif value < 2500:
        mode = "MODE 2"
        led2.value(1)

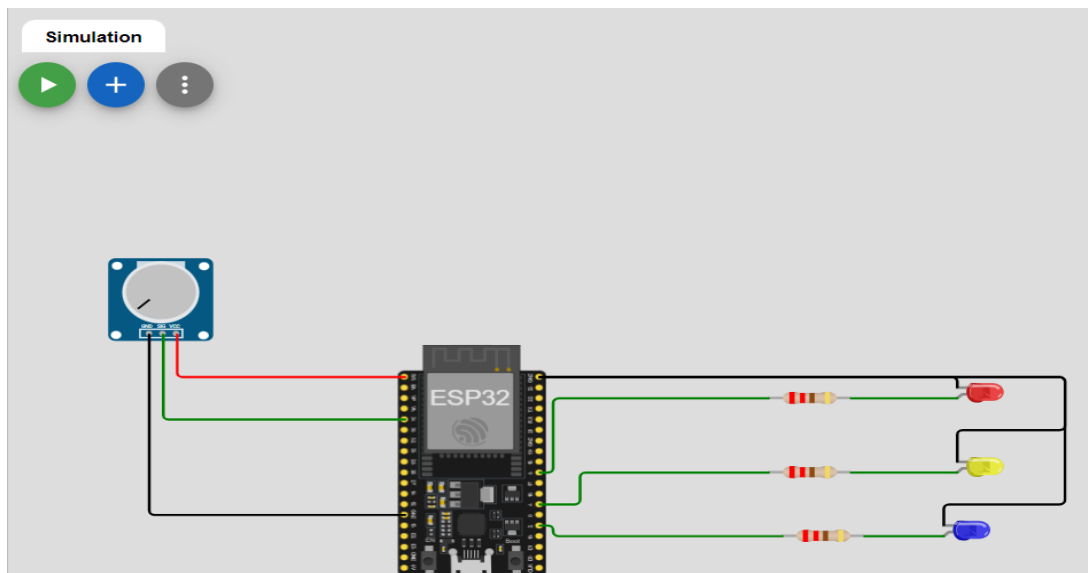
    else:
        mode = "MODE 3"
        led3.value(1)
```

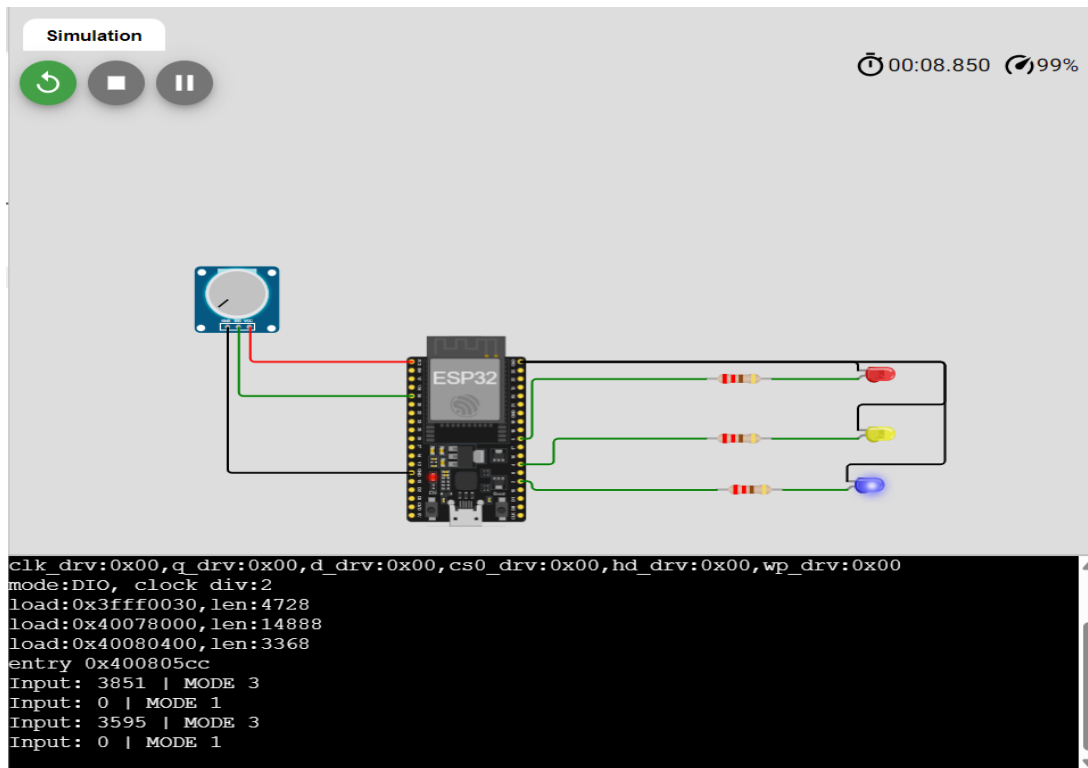
```
# Print only when mode changes
if mode != last_mode:
    print("Input:", value, "|", mode)
    last_mode = mode

time.sleep(0.2)
```

Output: The system indicates the recommended crop (such as Rice, Maize, or Cotton) using LEDs based on the input values. Different LEDs represent different crop outputs, and the output updates dynamically as the input values are varied. This demonstrates real-time crop prediction using the embedded TinyML logic.

Photos/ Screenshot/Snapshot :





Outcome: The project successfully demonstrates a TinyML-based crop recommendation system that performs real-time predictions on an embedded device. It enables efficient and data-driven crop selection using LED indicators for output, showcasing the practical application of IoT and machine learning in sustainable agriculture.

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